Question		Answer	Marks	AO	Guidance
11	(a)	**DR** $f(x) = (x+2)(x-1)(x-6)$	**M1**	**1.1a**	Attempt to factorise the quadratic factor or solve the equation $x^2 - 7x + 6 = 0$ algebraically
		$x = 1, 6$	**A1**	**1.1**	Must follow algebraic method
		$x = -2$	**B1**	**1.1**	Allow if seen
			[3]		
11	(b)	**DR** $y = x^3 - 5x^2 - 8x + 12$			
		$\frac{dy}{dx} = 3x^2 - 10x - 8$	**M1**	**2.1**	Differentiating their expanded $f(x)$
		When $x = 4$, $\frac{dy}{dx} = 48 - 40 - 8 = 0$ so stationary point	**M1**	**2.2a**	Substitution seen to give zero Also allow evidence of solving their $\frac{dy}{dx} = 0$ (eg factors, formula or both roots seen) Also allow for max or min stated
		$\frac{d^2y}{dx^2} = 6x - 10$	**M1**	**2.1**	Attempt to differentiate their first derivative
		When $x = 4$, $\frac{d^2y}{dx^2} [= 14] > 0$ so minimum point	**A1**	**2.2a**	Draw a correct conclusion from correct value
		Alternative method			
		When $x = 3$, $\frac{dy}{dx} = -11$ When $x = 5$, $\frac{dy}{dx} = 17$	**M1**		Evaluates gradient for points $-\frac{2}{3} < x < 4$ and for $x > 4$
		Gradient changes from negative to positive so minimum point	**A1**		Draw a correct conclusion from correct values
			[4]		

Question		Answer	Marks	AO	Guidance
11	(c)	**DR** Solving $\frac{dy}{dx} = 3x^2 - 10x - 8 = 0$ gives $x = -\frac{2}{3}$	**B1**	**1.1**	Correct exact x-coordinate found.
		$y = \frac{400}{27}$	**B1** **[2]**	**1.1**	Exact y-coordinate found. Allow for only for fraction or recurring decimal $14.\dot{8}1\dot{4}$